Adaptive Reference-Related Graph Embedding for Hyperspectral Anomaly Detection

Yubo Ma, Siyu Cai, and Jie Zhou

***Abstract*— Graph embedding (GE) provides an effective way to reveal the intrinsic feature of high-dimensional data on the foundation of preserving topological properties. Under the framework of GE, the hyperspectral image can be represented by a weighted graph, where pixels and similarities among them are treated as vertices and edge weights, respectively. In this article, an adaptive reference-related GE (ARGE) method is proposed to efficaciously obtain the low-dimensional feature and improve computational efficiency. The ARGE method is composed of two primary processes. The key to connecting these two processes is the reference vertices set, which is the abstraction of graph topological features. First, the reference vertices are adaptively selected through a three-step adaptive reference set selection (ARSS) algorithm. Second, the original high-dimensional graph is embedded as a low-dimensional graph through preserving the reference-related structure. Specifically, the pairwise similarities between vertices and reference vertices are preserved in embedding space. In addition, a new hybrid dissimilarity measure of Rao distance and spectral information divergence (RD-SID) is designed to depict the spectral difference between pixels. To evaluate the effectiveness of the proposed method, the obtained low-dimensional feature is fed into the anomaly detector to detect anomalous pixels. The experimental results on five real and one synthetic hyperspectral datasets demonstrate the superiority of the proposed ARGE method over the compared feature extraction methods.**

***Index Terms*— Anomaly detection, dissimilarity measure, graph embedding (GE), reference vertices, spectral clustering.**

I. Introduction

HYPERSPECTRAL image (HSI), differing from conventional image, is endowed with abundant spectral information. Therefore, the HSI possesses tremendous potential to be exploited in distinguishing different objects. This superior property allows hyperspectral anomaly detection to achieve excellent performance in a wide range of applications.

Compared with target detection, which relies on prior knowledge of target, anomaly detection is not supposed to have any information about potential targets [1], [2]. One of the common interpretations for anomaly is a variation from the normal surroundings [3], [4]. As for HSI, supposing that the unknown background and anomalous targets do exist objectively, anomaly detection takes the majority of pixels with similar spectra as background while treating those pixels with significant spectral discrepancy from the background as anomalies.

Among all the anomaly detection algorithms, one of the well-known benchmarks is the Reed–Xiaoli (RX) detector [5]. On the premise that the background follows a Gaussian distribution, the RX detector is essentially developed based on a generalized likelihood ratio test. On the foundation of the RX, some improved variants are proposed, such as local RX (LRX) [6] and kernel RX (KRX) [7]. These RX-based methods can be summarized into a distribution-based category, while the low-rank and sparse matrix decomposition (LRaSMD) method [8] can be concluded into a reconstruction-based category. The LRaSMD-based Mahalanobis distance for hyperspectral anomaly detection (LSMAD) [9] reconstructs an HSI matrix into the sum of a low-rank matrix and a sparse one and then builds a Mahalanobis distance-based anomaly detector. The graph and total variation regularized low-rank representation (GTVLRR) model [10] preserves the intrinsically local manifold structure and neighboring spatial homogeneity. To fit different types of anomalies and noise, the LRaSMD-based model incorporating the mixture of Gaussian (LSDM-MoG) [11] applies the Gaussian mixture model into the sparse component. In order to dig out more information of anomalies contained in the low-rank component, the LSDM with density peak guided collaborative representation (LSDDPCRD) [12] first adopts the density peak value to get a purified background set and constructs a low-rank collaborative-representation-based detector (CRD) and then obtains a weighted fusion result of low-rank detection and sparse one. Besides, to make better use of spatial association of HSI, an attempt of replacing matrix decomposition with tensor decomposition has been made. The prior-based tensor approximation (PTA) [13] decomposes HSI into a background tensor and an anomaly tensor through integrating low-rank, sparse, and piecewise-smooth priors. The total variation regularized low-rank tensor decomposition and collaborative representation (LRTDCRD) [14] separates an HSI into the background and mixed information components via total variation regularized low-rank tensor decomposition (LRTD) model and detects anomalies by fusing the detection results on the two components, which are obtained based on domain transform recursive filter and CRD.

The rich spatial and spectral information contained in HSI provides great help for the abovementioned anomaly detectors to achieve excellent performances even with limited prior information. Unfortunately, it is always counter-productive to have excessive information and spectral resolution. The

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high dimension of spectra may give rise to Hughes phenomenon [15], which manifests as initially rising and then gradually decreasing of detection accuracy along with the continuously increasing of dimension. Therefore, reducing redundant dimension while preserving sufficient information is an efficient way to avoid the Hughes phenomenon [16], [17], [18].

With dimensionality reduction (DR) gaining wider concern, previous work demonstrates that DR is an effective technique to eliminate redundant information and therefore enhances the overall performance of detection and classification [19], [20], [21]. The DR of HSI can be roughly divided into two patterns: feature extraction (FE) and band selection (BS). The former obtains a low-dimensional feature through a mapping process, while the latter selects a desired subset of bands directly [22].

As for BS, the maximum-variance principal component analysis (MVPCA) [23] is a classical ranking-based method, which chooses bands according to their variances. For the sake of eliminating the impact of noise, some clustering-based methods are proposed. The enhanced fast density-peak-based clustering (E-FDPC) [24] selects the bands that are the cluster centers with high local density. However, the cluster center may not possess the highest discrimination. To tackle this, the normalized cut-based optimal clustering with MVPCA (NC-OC-MVPCA) [25] picks those high-discriminative bands from each cluster based on spectral clustering. In addition to the clustering-based methods, the optimal neighborhood reconstruction (ONR) [26] chooses a band in the light of its reconstruction ability and involves a noise identification mechanism to avoid selecting noisy bands.

In terms of FE, the traditional statistical methods, such as principal component analysis (PCA) [27] and linear discriminant analysis (LDA) [28], rely on the assumption that both background and target are relatively simple and background obeys a multivariate Gaussian distribution. However, a real HSI is more complex and possesses the low-dimensional manifold structure embedded in high-dimensional Euclidean space [29], [30]. For the sake of revealing the intrinsic structure and nonlinear characteristic of HSI, three epoch-making manifold learning methods are proposed, including isometric mapping (ISOMAP) [31], locally linear embedding (LLE) [32], and Laplacian eigenmap (LE) [33]. Although they have great performances in nonlinear FE, the above methods all get implicit mappings. In other words, it is infeasible for them to directly learn a low-dimensional representation of out-of-sample data. To obtain the explicit projection, several linear algorithms are then put forward, such as neighborhood preserving embedding (NPE) [34], locality preserving projection (LPP) [35], and linear local tangent space alignment (LLTSA) [36].

The aforementioned manifold learning methods can be unified under the framework of graph embedding (GE) [37]. The essence of GE is finding the low-dimensional feature representation of high-dimensional data under the graph structure [38]. Taking an HSI as a graph, each pixel is a high-dimensional vertex and the similarity between pixels forms an edge. In a complex graph, the connections among vertices are also intricate, which could be direct or indirect, and simple or multiple. To depict the complicated relationship, the transition probability in random walk model has been utilized as the similarity measure between vertices under the framework of statistical relational learning [39]. The t-distributed stochastic neighbor embedding (t-SNE), one of the most popular DR methods with powerful visualization effect, aims at minimizing the overall mismatch between conditional transition probabilities in the original and embedding spaces [40]. However, the low computational efficiency makes t-SNE weak in large-scale graph processing and the existence of low-dimensional crowding problem makes it inapplicable to an intrinsically high-dimensional graph. In order to cope with this problem, the large-scale information network embedding (LINE) [41] presents a structure-preserving network embedding aspect, where both first- and second-order approximations are considered. On this foundation, GraRep [42] captures the higher order relational properties and reveals global structural information of the graph. To extract more discriminative features from graph structure, the neighbor, tangential, and statistical properties are jointly utilized in the multistructure unified discriminative embedding (MUDE) [43].

Despite many excellent properties of GE framework, previous GE methods neglect the fact that noise is always inevitable. To amply dig up information from spatial as well as spectral domain and eliminate the influence of noise, orthogonal total variation component analysis (OTVCA) [44] is proposed to seek low-dimensional features through a spatial-regularized and low-rank model, where total variation is used as penalty to promote local smoothness. The OTVCA is adept in capturing features of HSI with high signal-to-noise ratio (SNR). Recently, some superpixel-based FE methods are proposed to make full use of global–local and spatial–spectral information. A joint adaptive structure density (JASD)-based method [45] preserves both local structural information and global edge details through density-assisted mean operation and recursive filtering. Spectral–spatial and superpixel-wise PCA (S^3-PCA) [30] separately extracts the features from the whole HSI and homogeneous superpixels via PCA and then concatenates the global and local features in accordance with the spatial location. The common ground of these methods can be summarized as performing local smoothing. On the one hand, smoothing can effectively eliminate the influence of noise. On the other hand, it could lead to the blurring of edge. In a real HSI anomaly detection scene, the anomalous pixels might be distributed scatteredly and irregularly. In this case, the subregion smoothing methods could be inadequate for the detection of micro and isolated anomalies.

In this article, an intrinsic FE method, termed adaptive reference-related GE (ARGE), is proposed. Fundamentally, the ARGE is a structure-preserving GE method. Due to the low computational efficiency of preserving global structure, many conventional structure-preserving methods are inclined to preserve local neighborhood structure [46]. They ensure that those adjacent vertices in high-dimensional space also keep close in low-dimensional space. However, they do not guarantee that the distant vertices in the original space remain

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far away from each other in the embedding space. Motivated by the above limitation, a relative structure-preserving strategy is considered in this work. The proposed ARGE adaptively selects some representative reference vertices through three steps. The selected reference vertices reflect the global and local topological characteristics. Then, the ARGE carries out GE on the foundation of preserving the relative graph structure related to reference vertices.

The distinctive contributions of this article are listed as follows.

1) A hybrid dissimilarity measure of Rao distance and spectral information divergence (RD-SID) is designed to depict the spectral difference between pixels. It turns out that RD-SID has a better power of relative spectral discriminability (PRSD) than some classical distance metrics.
2) An adaptive reference set selection (ARSS) algorithm is introduced to select global reference vertices from local subgraphs. This process takes into account both local and global graph structure information and extracts the intrinsic features of graph. The selected reference set maintains the "shape" of graph.
3) The embedded dimension in the proposed ARGE method relies on the number of reference vertices, which indicates that there is no need to preset a subjective dimension. Meanwhile, the reference-related structure-preserving GE method can greatly improve computational efficiency than global structure-preserving methods.

The rest of this article is organized as follows. Section II briefly reviews some related work. Section III introduces the representative point selection strategy and proposes the ARGE method to extract the low-dimensional feature of hyperspectral data. In Section IV, the ARGE is experimentally evaluated in a series of real HSI experiments as well as synthetic one. Finally, Section V gives the conclusions.

II. Related Work

Throughout this article, some frequently used notations are given. A raw hyperspectral dataset is denoted as a matrix $\mathbf{Z} = [\boldsymbol{z}_1, \boldsymbol{z}_2, \ldots, \boldsymbol{z}_n]^T \in \mathbb{R}^{n\times D}$, where n and D are the total numbers of pixels and bands, respectively, the D-dimensional spectral vector $\boldsymbol{z}_i$ $(i = 1, 2, \ldots, n)$ represents the ith pixel, and z_{il} is the lth spectral value of $\boldsymbol{z}_i$. Applying sum normalization, the normalized hyperspectral dataset is denoted as $\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n]^T \in \mathbb{R}^{n\times D}$, where the ith normalized spectral vector $\boldsymbol{x}_i = \boldsymbol{z}_i / \sum_{l=1}^{D} z_{il}$. The reference vertices matrix $\mathbf{R} = [\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_d]^T \in \mathbb{R}^{d\times D}$ is composed of d representative vertices. $\mathbf{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n]^T \in \mathbb{R}^{n\times d}$ is a normalized low-dimensional feature representation of $\mathbf{X}$, where the embedding dimension is d $(d \ll D)$. In this article, the embedding dimension and the number of reference vertices are equal despite their different physical meanings.

A. Dissimilarity Measure

From the perspective of spectral similarity, there always exists compactness among same-class pixels and separability among different-class pixels. However, atmospheric noise and measurement error lead to the increase of intraclass variation and the decrease of interclass discrepancy. To mitigate the impact, an appropriate metric is expected, which could capture both "shape" and "magnitude" of spectra and magnify similarities among homogeneous pixels as well as dissimilarities among heterogenous ones.

There are some extensively used distances in HSI processing, such as Euclidean distance (ED) [47], spectral angle mapper (SAM) [45], and spectral information divergence (SID) [48]. Each metric possesses unique property. The ED concentrates on the absolute reflectance intensity, while SAM focuses on a relative spectral angle and is invariant to the scaling of reflectance intensity. However, the ED and SAM only depict linear dissimilarity. In order to quantify nonlinear spectral discrepancies from different perspectives, two basic dissimilarity measures, Rao distance (RD) [49] and SID, have been taken into consideration in high-dimensional space. The RD and SID are defined, respectively, as

$$d_{\mathrm{RD}}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \left(\sum_{l=1}^{D}(\sqrt{x_{il}} - \sqrt{x_{jl}})^2\right)^{\frac{1}{2}} \tag{1}$$

$$d_{\mathrm{SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \sum_{l=1}^{D}\left(x_{il}\ln\frac{x_{il}}{x_{jl}} + x_{jl}\ln\frac{x_{jl}}{x_{il}}\right) \tag{2}$$

where $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ are two normalized D-dimensional vectors.

On the one hand, taking an information-theoretical approach, the SID can effectively reflect spectral dissimilarity via bidirectional relative entropy and is demonstrated more superiority than other distances [50]. On the other hand, as found in previous studies, high-dimensional hyperspectral data possess low-dimensional manifold structure [51]. Different from Euclidean space, the shortest path joining two points on a manifold, which is termed geodesic line, is curved instead of plane. In this case, the RD can be applied to depict the intrinsic geodesic distance between two hyperspectral pixels. To draw on the merits of RD as well as SID and let them complement each other, an intuitive idea is to integrate two metrics. Therefore, a hybrid dissimilarity measure RD-SID is formed by the combination of RD and SID, which is given as

$$d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \tan(d_{\mathrm{RD}}(\boldsymbol{x}_i, \boldsymbol{x}_j)) \cdot d_{\mathrm{SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{3}$$

where $d_{\mathrm{RD}}(\boldsymbol{x}_i, \boldsymbol{x}_j)$ and $d_{\mathrm{SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j)$ are given by (1) and (2), respectively.

The tangent function is used to give a weight on SID [48], [52], [53], [54]. Also, it is guaranteed that $\tan(d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j))$ increases monotonously with respect to $d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{x}_j)$. The RD-SID actually makes a tradeoff between geodesic distance and information divergence. A great RD-SID shows an obvious difference on both RD and SID, while a tiny RD-SID indicates strong similarity. The RD-SID presents the tremendous potential of spectral discrimination in high-dimensional space. However, using RD-SID is confronted with zero-denominator problem when calculating low-dimensional reference-vertex-related distance. Under such circumstances, the RD-SID is not a suitable dissimilarity measure in the low-dimensional embedding space. In the ARGE method,

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Bhattacharyya distance (BD) is used as the dissimilarity measure of low-dimensional features because of its excellent properties [55], [56], [57], [58]. After isometric square-root transformation, a feature vector can be treated as a point on a hypersphere, more precisely, the nonnegative quadrant of a unit hypersphere. Then, the spherical distance on a hypersphere can be effectively calculated through

$$d_{\mathrm{BD}}(\boldsymbol{x}_i, \boldsymbol{x}_j) = -\ln \sum_{l=1}^{D} \sqrt{x_{il} x_{jl}}. \tag{4}$$

B. Graph Spectral Clustering

A weighted graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$ is composed of a vertex set $\mathcal{V}$, an edge set $\mathcal{E}$, and a weight matrix $\mathbf{W}$. The weight between vertices $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ is w_{ij}. Each vertex has a degree $d_i = \sum_{j=1}^{n} w_{ij}$. The diagonal matrix $\mathbf{D} = \mathrm{diag}(\{d_i\}_{i=1}^{n})$ is called the degree matrix. The Laplacian matrix, as one of the most common matrix representations of graph, is defined as $\mathbf{L} = \mathbf{D} - \mathbf{W}$. The spectrum of graph $\mathcal{G}$ can be deemed as a feature of $\mathcal{G}$, which is invariant to graph isomorphism [59].

Graph spectral clustering, on the foundation of graph spectrum, is more like segmentation than aggregation. It aims at finding an optimal way to divide the graph $\mathcal{G}$ into K disjoint subgraphs $\mathcal{G}_k (k = 1, \ldots, K)$, which is minimizing the loss of graph cut. One of the classical spectral clustering algorithms is normalized cut (Ncut) [60]. Considering the volume equilibrium of subgraphs, the loss of Ncut is defined as

$$\mathrm{Loss}_{\mathrm{Ncut}} = \frac{1}{2} \sum_{k=1}^{K} \frac{W(\mathcal{V}_k (\mathcal{V}_k)^{\mathrm{c}})}{\mathrm{vol}(\mathcal{V}_k)} \tag{5}$$

where $(\mathcal{V}_k)^{\mathrm{c}}$ is the complementary set of the kth subset $\mathcal{V}_k$, $W(\mathcal{V}_k, (\mathcal{V}_k)^{\mathrm{c}}) = \sum_{\boldsymbol{v}_i \in \mathcal{V}_k, \boldsymbol{v}_j \in (\mathcal{V}_k)^{\mathrm{c}}} w_{ij}$ is the sum of edge weights between $\mathcal{V}_k$ and $(\mathcal{V}_k)^{\mathrm{c}}$, and $\mathrm{vol}(\mathcal{V}_k) = \sum_{\boldsymbol{v}_i \in \mathcal{V}_k} d_i$ is the volume of $\mathcal{V}_k$.

The Ncut loss (5) can be rewritten as $\mathrm{tr}(\mathbf{H}^T \mathbf{L} \mathbf{H})$ with the condition $\mathbf{H}^T \mathbf{D} \mathbf{H} = \mathbf{I}$. The scaled incidence matrix $\mathbf{H}$ depicts the relationship between vertices and subgraphs, in which the entry is denoted as $h_{ik} = (1/\sqrt{\mathrm{vol}(\mathcal{V}_k)}) \mathbb{1}_{\{\boldsymbol{v}_i \in \mathcal{V}_k\}}$, where $\mathbb{1}$ is the indicator function.

Let $\mathbf{F} = \mathbf{D}^{1/2} \mathbf{H}$ and $f_{ik} = \sqrt{d_i} h_{ik}$ be the entry of $\mathbf{F}$. Then, the minimization of Ncut loss in (5) can be formulated as the following optimization problem:

$$\begin{aligned} &\min_{\mathbf{F} \in \mathbb{R}^{n \times K}} \mathrm{tr}(\mathbf{F}^T \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2} \mathbf{F}) \\ &\text{s.t. } \mathbf{F}^T \mathbf{F} = \mathbf{I} \\ &\qquad f_{ik} \in \{0, c_{ik}\}, \quad i = 1, \ldots, n, \ k = 1, \ldots, K \end{aligned} \tag{6}$$

where $c_{ik} = \sqrt{d_i / \sum_{\boldsymbol{v}_i \in \mathcal{V}_k} d_i} \in (0, 1]$.

Unfortunately, the above optimization problem is NP-hard under the discrete constraint on $\mathbf{F}$ [61]. Adopting the well-known relaxation technique to replace the constrains $f_{ik} \in \{0, c_{ik}\}$ by $f_{ik} \in [0, 1]$, a solution of such relaxed problem of the optimization problem (6) is the K eigenvectors of $\mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$ corresponding to the first K eigenvalues in nondescending order. Finally, every vertex gets an approximate discrete cluster label via K-means clustering.

III. Methodology

A. Adaptive Reference Set Selection

The concept of reference vertex plays a crucial role throughout this article. The reference vertices reveal the intrinsic features of graph and are the abstraction of locally dense communities in a sparse graph. An appropriate reference vertices set should be representative and independent. The representativeness means reference vertices could maintain the "shape" of graph and eliminate redundant information. The independence signifies reference vertices should be distant from each other and have no relevance. Accordingly, the selection of reference vertices must meet these two requirements.

In this article, an ARSS algorithm is proposed to adaptively seek reference vertices of a graph on the foundation of graph spectral clustering. Different from the traditional spectral clustering, the ARSS is independent of any prior information, such as the quantity of clusters. To estimate the number of clusters, the inflection point with maximal convexity in nondescending eigenvalues needs to be found. Thus, we design a relative eigengap maximization criterion based on [62], where the objective function has a form of relative second-order difference. Under this criterion, the optimal number of clusters $\hat{K}$ can be estimated by

$$\hat{K} = \underset{k \in \{2, \ldots, n-1\}}{\arg\max} \frac{\lambda_{(k+1)} - \lambda_{(k-1)}}{\lambda_{(k)}} \tag{7}$$

where $\lambda_{(k)}$ denotes the kth smallest eigenvalue.

$\hat{K}$ can be deemed as the intrinsic dimensionality of data. Also, data can be divided into $\hat{K}$ categories via K-means clustering. After clustering, there always exists the demand of seeking the representative vertex or centroid of each cluster. The cluster centroids are close to their within-cluster vertices but are not guaranteed to be far away from between-cluster vertices, especially on nonconvex data.

Therefore, a representative vertices selection strategy is proposed to select those quintessential vertices from different clusters, which have small differences with their farthest within-cluster vertices and large gaps with their closest different-cluster vertices.

Taking a pixel as a vertex, the hyperspectral data $\boldsymbol{x}_i \in \mathbb{R}^D (i = 1, \ldots, n)$ form a vertex set $\mathcal{V}$. Suppose that $\mathcal{V}$ can be divided into K clusters $\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_K$. For the ith vertex in the kth cluster $\boldsymbol{v}_{ki}$, the distances between $\boldsymbol{v}_{ki}$ and its within-cluster vertices form

$$\mathcal{D}_w(\boldsymbol{v}_{ki}) = \left\{ d_{\text{RD-SID}}(\boldsymbol{v}_{ki}, \boldsymbol{v}_j) \cdot \mathbb{1}_{\{\boldsymbol{v}_{ki} \in \mathcal{V}_k, \boldsymbol{v}_j \in \mathcal{V}_k\}} \right\}_{j=1}^{n_k} \tag{8}$$

and the distances between $\boldsymbol{v}_{ki}$ and its between-cluster vertices compose

$$\mathcal{D}_b(\boldsymbol{v}_{ki}) = \left\{ d_{\text{RD-SID}}(\boldsymbol{v}_{ki}, \boldsymbol{v}_j) \cdot \mathbb{1}_{\{\boldsymbol{v}_{ki} \in \mathcal{V}_k, \boldsymbol{v}_j \notin \mathcal{V}_k\}} \right\}_{j=1}^{n-n_k} \tag{9}$$

where n_k and n are the numbers of vertices in $\mathcal{V}_k$ and $\mathcal{V}$, respectively. Then, the representative vertex of the kth cluster $\tilde{\boldsymbol{v}}_k$ is obtained through

$$\tilde{\boldsymbol{v}}_k = \underset{\boldsymbol{v}_{ki} \in \mathcal{V}_k}{\arg\min} \left\{ \frac{\max \mathcal{D}_w(\boldsymbol{v}_{ki})}{\min \mathcal{D}_b(\boldsymbol{v}_{ki})} \right\}_{i=1}^{n_k}. \tag{10}$$

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Algorithm 1 ARSS

Input: The HSI data $\mathbf{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n]^T \in \mathbb{R}^{n\times D}$, the number of sub-regions after partitioning M.

Output: Reference vertices $\mathbf{R} = [\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_d]^T \in \mathbb{R}^{d\times D}$.

Begin:

1: Estimate background spectrum $\hat{\boldsymbol{b}} \leftarrow \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i$;
2: Segment $\mathbf{X}$ into $\{\mathbf{X}^m\}_{m=1}^{M}$ by sliding window;
3: **for** $m = 1 \rightarrow M$ **do**
4: $\quad \tilde{\boldsymbol{x}}_{2m-1} \xleftarrow{\text{Max pooling}} \arg\max_{\boldsymbol{x}_i^m \in \mathbf{X}^m} \{d_{\text{RD-SID}}(\boldsymbol{x}_i^m, \hat{\boldsymbol{b}})\}_{i=1}^{n/M}$;
5: $\quad \tilde{\boldsymbol{x}}_{2m} \xleftarrow{\text{Min pooling}} \arg\min_{\boldsymbol{x}_i^m \in \mathbf{X}^m} \{d_{\text{RD-SID}}(\boldsymbol{x}_i^m, \hat{\boldsymbol{b}})\}_{i=1}^{n/M}$;
6: **end for**
7: $\widetilde{\mathbf{X}} = [\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_{2M}]^T$;
8: Form the vertex set $\mathcal{V}$, compute the weight matrix $\mathbf{W}$ according to (11) and construct the graph $\mathcal{G}$ for $\widetilde{\mathbf{X}}$;
9: $\mathbf{L}, \mathbf{D} \leftarrow \mathbf{W}$;
10: $\boldsymbol{\lambda}, \mathbf{F} \leftarrow \text{EigenDecomposition}(\mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2})$;
11: $d \leftarrow \arg\max_{a\in\{2,\ldots,2\ M-1\}} \frac{\lambda_{(a+1)}-\lambda_{(a-1)}}{\lambda_{(a)}}$;
12: $\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_d \leftarrow \text{KMeans}(\mathbf{F}, d)$;
13: **for** $k = 1 \rightarrow d$ **do**
14: $\quad \boldsymbol{r}_k \leftarrow \arg\min_{\boldsymbol{v}_{ki}\in\mathcal{V}_k} \left\{ \frac{\max \mathcal{D}_w(\boldsymbol{v}_{ki})}{\min \mathcal{D}_b(\boldsymbol{v}_{ki})} \right\}_{i=1}^{n_k}$;
15: **end for**
16: $\mathbf{R} = [\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_d]^T$;
17: $diam(\mathbf{R}) = \max\{\{d_{\text{RD-SID}}(\boldsymbol{r}_i, \boldsymbol{r}_j)\}_{j=1}^{d}\}_{i=1}^{d}$;
18: **while** $\max\{\min\{d_{\text{RD-SID}}(\tilde{\boldsymbol{x}}_i, \boldsymbol{r}_j)\}_{j=1}^{d}\}_{i=1}^{2M} \geq diam(\mathbf{R})$ **do**
19: $\quad \boldsymbol{r}_{d+1} \leftarrow \arg\max_{\tilde{\boldsymbol{x}}_i\in\widetilde{\mathbf{X}}} \{\min\{d_{\text{RD-SID}}(\tilde{\boldsymbol{x}}_i, \boldsymbol{r}_j)\}_{j=1}^{d}\}_{i=1}^{2M}$;
20: $\quad \mathbf{R} = [\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_d, \boldsymbol{r}_{d+1}]^T$;
21: $\quad diam(\mathbf{R}) = \max\{d_{\text{RD-SID}}(\boldsymbol{r}_{d+1}, \boldsymbol{r}_j)\}_{j=1}^{d}$;
22: $\quad d = d + 1$;
23: **end while**

End

After determining the number of clusters and developing the representative vertices selection strategy, the ARSS can be constructed. The complete procedures of ARSS are summarized in Algorithm 1 and the schematic of ARSS is shown in Fig. 1. Generally, there are three primary steps in the ARSS. The first step screens out some representative pixels from HSI and extracts local features, whereas the second step selects reference vertices and characterizes global shape. Then, the third step adds eligible vertices into reference set and supplements extra information. The reference vertices selected through the second and third steps characterize background and anomaly information and provide assistance for subsequent anomaly detection. The three steps are corresponding to the 1st to 7th, 8th to 16th, and 17th to 23rd operations in Algorithm 1. The ARSS process is elaborated as follows.

1) *First Step:* In order to reduce computational complexity, the downsampling step is designed to carry out data reduction while retaining the local features of HSI. This process uses the idea of the divide-and-conquer algorithm, which can be described as follows.
   a) *Divide:* The original HSI data $\mathbf{X} \in \mathbb{R}^{n\times D}$ are divided into M subregion HSIs $\{\mathbf{X}^m\}_{m=1}^{M}$ by sliding window.

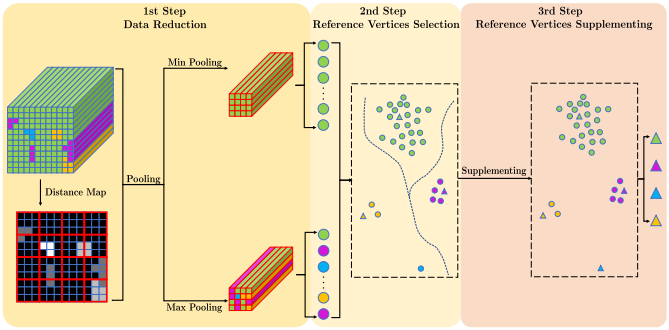

Fig. 1. Schematic of the ARSS algorithm.

   b) *Conquer:* For each sub-HSI $\mathbf{X}^m \in \mathbb{R}^{(n/M)\times D}$, two anchor pixels with the largest and smallest distance from the estimated background pixel are selected through the "max" and "min" pooling, which are deemed as the locally distinctive features of HSI.
   c) *Combine:* Finally, $2M$ anchor pixels after pooling are merged into a new matrix of representative pixels $\widetilde{\mathbf{X}} = [\tilde{\boldsymbol{x}}_1, \tilde{\boldsymbol{x}}_2, \ldots, \tilde{\boldsymbol{x}}_{2M}]^T \in \mathbb{R}^{2M\times D} (2M \ll n)$.
2) *Second Step:* The significance of this step is to further extract reference vertices from the representative vertices through spectral clustering. First, the $2M \times 2M$ weight matrix $\mathbf{W} = (w_{ij})$ is constructed based on $\widetilde{\mathbf{X}}$, where an entry w_{ij} is given by

$$w_{ij} = \exp\left(-\frac{d_{\text{RD-SID}}(\tilde{\boldsymbol{x}}_i, \tilde{\boldsymbol{x}}_j)}{\sigma_w}\right) \tag{11}$$

   where $\sigma_w = (1/(2M)^2)\sum_{i,j} d_{\text{RD-SID}}(\tilde{\boldsymbol{x}}_i, \tilde{\boldsymbol{x}}_j)$. Then, the corresponding normalized Laplacian matrix $\mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ is obtained. Afterward, the eigenvalue vector $\boldsymbol{\lambda} \in \mathbb{R}^{2M}$ and the eigenvector matrix $\mathbf{F} \in \mathbb{R}^{2M\times 2M}$ are acquired via eigendecomposition. The clustering number d is estimated based on the relative eigengap maximization criterion. After K-means clustering, vertices are divided into d clusters. Then, d reference vertices $\{\boldsymbol{r}_k\}_{k=1}^{d}$ are selected through adopting the representative vertices selection strategy.
3) *Third Step:* The third step is a supplementary step, where qualified unselected vertices will be added to the reference set. Specifically, the maximum distance between any two reference vertices is taken as the diameter of reference set. Then, calculate the distance between each representative vertex $\tilde{\boldsymbol{x}}_i$ and its nearest reference vertex. If the maximum distance is greater than the diameter, the corresponding representative vertex is added to the reference set. Otherwise, the reference set remains unchanged. Repeat the above procedure until no distance is greater than the diameter.

B. Adaptive Reference-Related GE

GE is a mapping process to obtain the compressed representation of high-dimensional graph. In the embedding process, the invariants are vertices and edges themselves, while the

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variation is the dimension of vertex. The bridge connecting both sides is the adjacent similarity. To ensure the availability of embedding process, the adjacent similarity in the embedding space should approach to that in the original space. In other words, an excellent GE algorithm is supposed to minimize the difference between the before- and after-embedding adjacent similarity matrices.

However, how to define "adjacency" or how to find "neighbor" has always been a problem. Traditionally, there are three strategies to construct an adjacency matrix, including k-nearest neighbor, ϵ-neighborhood, and full connection. Constructing adjacency matrix through the first two strategies relies heavily on hyperparameter tuning. The full connection strategy, though not depending on any hyperparameter, leads to low computational efficiency. To reduce computation without relying on hyperparameter, a special graph structure is considered, termed complete bipartite graph.

Consider a complete bipartite graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{P}\}$. The bipartition indicates that the vertex set $\mathcal{V}$ can be divided into two disjoint subsets $\mathcal{V}_1$ and $\mathcal{V}_2$ [63], [64], which consist of n pixels $\boldsymbol{x_i}$ $(i = 1, \ldots, n)$ and d reference vertices $\boldsymbol{r_j}$ $(j = 1, \ldots, d)$. The completeness means that for any two elements $\boldsymbol{x}_i$ and $\boldsymbol{r}_j$, respectively, in $\mathcal{V}_1$ and $\mathcal{V}_2$, there always exists a connected edge. The collection of all edges is denoted as $\mathcal{E}$. For the adjacency matrix (also transition probability matrix) $\mathbf{P} \in \mathbb{R}^{n\times d}$, the entry p_{ij} represents the transition probability from $\boldsymbol{x}_i$ to $\boldsymbol{r}_j$.

The foundation of GE process is to select appropriate metric to measure the transition probability. As mentioned in Section II, the high-dimensional HSI data intrinsically lie on a manifold. The proposed RD-SID is adept in characterizing the geodesic distance on manifold, whereas it is inappropriate in the low-dimensional space. Therefore, the dissimilarity measures should be distinct in different space.

In the original high-dimensional space, for $i = 1, \ldots, n$ and $j = 1, \ldots, d$, the transition probability from a pixel $\boldsymbol{x}_i$ to a reference vertex $\boldsymbol{r}_j$ can be obtained using a Laplacian-kernel-based softmax function on RD-SID, which is defined as

$$p(\boldsymbol{x}_i, \boldsymbol{r}_j) = \frac{\exp\left(-\frac{d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{r}_j)}{\sigma}\right)}{\sum_{l=1}^{d} \exp\left(-\frac{d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{r}_l)}{\sigma}\right)} \tag{12}$$

where $\sigma = (1/(nd)) \sum_{i,j} d_{\text{RD-SID}}(\boldsymbol{x}_i, \boldsymbol{r}_j)$ is the scale parameter.

In the low-dimensional embedding space, $\boldsymbol{y}_i$ is the normalized low-dimensional feature of $\boldsymbol{x}_i$, which satisfies $\sum_{l=1}^{d} y_{il} = 1$, $i = 1, \ldots, n$. Also, $\tilde{\boldsymbol{r}}_j$ is the low-dimensional representation of $\boldsymbol{r}_j$, $j = 1, \ldots, d$. Taking BD as the dissimilarity measure, the transition probability from $\boldsymbol{y}_i$ to $\tilde{\boldsymbol{r}}_j$ can be defined as

$$q(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_j) = \frac{\exp\left(-\frac{d_{\text{BD}}(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_j)}{\tilde{\sigma}}\right)}{\sum_{l=1}^{d} \exp\left(-\frac{d_{\text{BD}}(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_l)}{\tilde{\sigma}}\right)} \tag{13}$$

where the scale parameter $\tilde{\sigma}$ is a constant and is taken as $(\log d)/2$ in this article.

Substituting (4) into (13), $q(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_j)$ is brought into a simplified form

$$q(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_j) = \frac{y_{ij}^{\frac{1}{2\tilde{\sigma}}}}{\sum_{l=1}^{d} y_{il}^{\frac{1}{2\tilde{\sigma}}}}, \quad i = 1, \ldots, n, \ \ j = 1, \ldots, d. \tag{14}$$

Algorithm 1 ensures the large pairwise distances among selected reference vertices in high-dimensional space. Therefore, a reasonable assumption is that a reference vertex only transfers to itself. More specifically, whether in high-dimensional or low-dimensional space, the transition probability between reference vertices should satisfy $p(\boldsymbol{r}_i, \boldsymbol{r}_j) = q(\tilde{\boldsymbol{r}}_i, \tilde{\boldsymbol{r}}_j) = \mathbb{1}_{\{i=j\}}$. Thus, $[\tilde{\boldsymbol{r}}_1, \tilde{\boldsymbol{r}}_2, \ldots, \tilde{\boldsymbol{r}}_d]^T \in \mathbb{R}^{d\times d}$ should be an identity matrix, which can be deemed as a set of orthonormal bases.

The embedding process of the ARGE relies on preserving a special graph structure. A traditional global structure-preserving method preserves the overall graph structure. Specifically, the edge set $\mathcal{E}$ and the corresponding weight matrix $\mathbf{W}$ should be consistent with the after-embedding edge set $\mathcal{E}^*$ and weight matrix $\mathbf{W}^*$. Also, the local structure-preserving method keeps the neighbor structure of each vertex. In order to efficiently maintain the global structure in the GE process, the ARGE method adopts a reference-related structure-preserving pattern. In this pattern, a desired GE process is to seek an optimal embedded graph $\mathcal{G}^* = \{\mathcal{V}, \mathcal{E}, \mathbf{Q}\}$ so that the loss from the transition probability matrix $\mathbf{P}$ to $\mathbf{Q}$ reaches the minimum. To measure the discrepancy from one probability distribution to another, the Kullback–Leibler divergence and cross entropy are used as two common loss functions. In our optimization problem, omitting the constant term, minimizing the Kullback–Leibler divergence degenerates into minimizing the cross entropy. Applying the cross entropy as the loss function, the reference-related structure-preserving process can be implemented by solving the following convex optimization problem with equality constraints:

$$\begin{aligned} &\min_{q_{ij}\geq 0} \sum_{i=1}^{n}\sum_{j=1}^{d} -p_{ij} \ln q_{ij} \\ &\text{s.t.} \sum_{j=1}^{d} q_{ij} = 1, \quad i = 1, \ldots, n \end{aligned} \tag{15}$$

where $p_{ij} = p(\boldsymbol{x}_i, \boldsymbol{r}_j)$ and $q_{ij} = q(\boldsymbol{y}_i, \tilde{\boldsymbol{r}}_j)$ are given by (12) and (13), respectively, and $0 \ln 0$ is defined to be 0.

According to [65], the cross-entropy loss reaches the minimum when $q_{ij} = p_{ij}$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, d$. Subsequently, using (14), we can easily obtain

$$y_{ij} = \left(p_{ij} \sum_{l=1}^{d} y_{il}^{\frac{1}{2\tilde{\sigma}}}\right)^{2\tilde{\sigma}}. \tag{16}$$

For $i = 1, \ldots, n$, from (16) and the fact that the vector $\boldsymbol{y}_i$ is normalized, we immediately have

$$\left(\sum_{l=1}^{d} y_{il}^{\frac{1}{2\tilde{\sigma}}}\right)^{2\tilde{\sigma}} = \frac{1}{\sum_{l=1}^{d} p_{il}^{2\tilde{\sigma}}} \tag{17}$$

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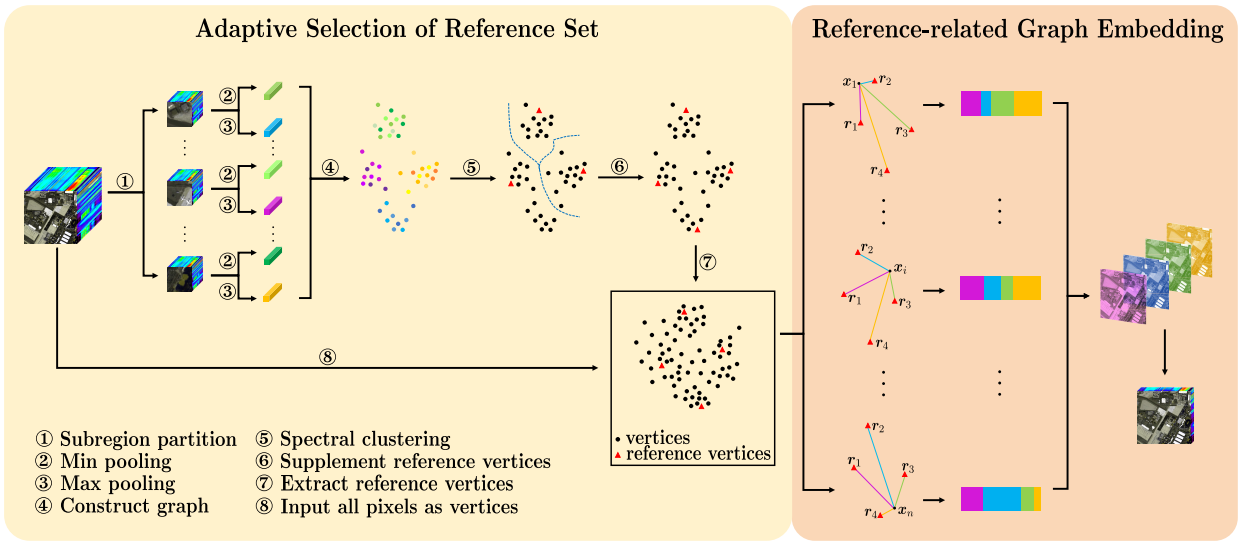

Fig. 2. Framework of the ARGE method.

and then

$$\boldsymbol{y}_i = \left[\frac{p_{i1}^{2\tilde{\sigma}}}{\sum_{l=1}^{d} p_{il}^{2\tilde{\sigma}}}, \frac{p_{i2}^{2\tilde{\sigma}}}{\sum_{l=1}^{d} p_{il}^{2\tilde{\sigma}}}, \ldots, \frac{p_{id}^{2\tilde{\sigma}}}{\sum_{l=1}^{d} p_{il}^{2\tilde{\sigma}}} \right]^T. \tag{18}$$

Finally, the obtained low-dimensional feature $\mathbf{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n]^T \in \mathbb{R}^{n \times d}$ is used as the input for successive anomaly detection. The overall framework of the proposed ARGE method is presented in Fig. 2.

IV. Experiments and Analysis

In this section, several experiments on five real HSIs and one synthetic scene are utilized to evaluate the effectiveness of the ARGE algorithm. All the experiments are tested on a Windows 10 Pro for Workstations 64-bit platform equipped with Intel Xeon Gold 6240M CPU (2.60-GHz) and 256-GB RAM.

A. Datasets Description

1) *San Diego Airport Dataset*: This dataset is a popular airborne HSI captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The entire image comprises 400 × 400 pixels with 224 spectral bands. After removing noisy bands of low SNR, there remain 189 available bands. In this article, two different regions of interest (ROIs) with a size of 100 × 100 and 200 × 200 are selected as the first and second real datasets, which are abbreviated as SD-I and SD-II. As shown in Fig. 3(a), three aircraft in both SD-I and SD-II are considered as abnormal objects, which consist of 57 and 134 anomalous pixels, respectively.
2) *Cri Dataset*: Another real dataset is the Cri dataset, which was collected by the Nuance Cri hyperspectral sensor. The Cri dataset covers an area of 400 × 400 pixels and has 46 spectral bands. In this HSI, meadow is considered as the background, while ten rocks are the anomalous objects to be detected, which contains 2216 anomalies [66]. The Cri dataset can be seen in Fig. 3(b).
3) *Gulfport Dataset*: This HSI was acquired by the AVIRIS sensor at Gulfport and obtained from the Airport–Beach–Urban dataset.[1] The spatial size of the image is 100 × 100 and the number of spectral channels is 191. The Gulfport HSI is shown in Fig. 3(c). In this dataset, 60 pixels are considered as anomalies [67].
4) *Cooke Dataset*: This dataset is provided from the project of Center for Imaging Science, Rochester Institute of Technology [68], and captured by HyMap sensor at a small town of Cooke City in Montana, USA.[2] The spatial size is 800 × 280 and the spectral dimension is 126. Fig. 3(e) shows the false-color image of the Cooke dataset, in which four kinds of vehicles are treated as anomalies.
5) *Synthetic Scene*: The synthetic image is generated based on the Cooke dataset. Following the previous work [69], [70], [71], the anomalous pixels are constructed based on the linear mixing model and then implanted on the 280 × 280 sub-HSI of the Cooke dataset. Specifically, the spectrum of a vehicle in the original HSI is selected as the anomalous target spectrum $\boldsymbol{t}$ and all 78 400 pure background pixels are denoted as $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_{78400}$. Then, the ith simulated anomalous pixel $\boldsymbol{s}_i$ is given as

$$\boldsymbol{s}_i = \beta \boldsymbol{t} + (1 - \beta)\boldsymbol{b}_i \tag{19}$$

where β is the mixing coefficient.
In order to characterize various types of anomalies, 25 anomalous targets with different intensities and shapes are implanted in the synthetic image. These anomalies are distributed in five rows and five columns. The mixing coefficient β is 0.05, 0.1, 0.2, 0.3, and 0.4 for different rows. In other words, anomalous targets possess equivalent anomalous intensity in the same row.

[1] http://xudongkang.weebly.com/datasets.html
[2] http://dirsapps.cis.rit.edu/blindtest/

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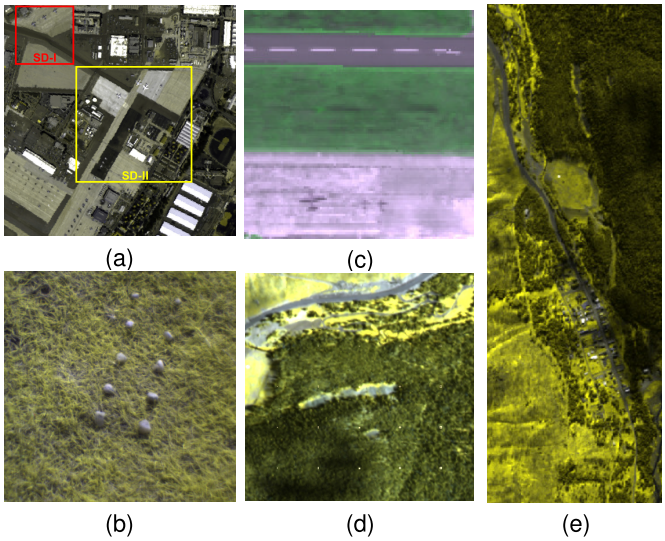

Fig. 3. Six HSIs. (a) SD-I and SD-II. (b) Cri. (c) Gulfport. (d) Synthetic scene. (e) Cooke.

TABLE I
OPTIMAL PARAMETERS SETTING OF DIFFERENT DR METHODS

	LLE	LLTSA	OTVCA	JASD	S^3-PCA	ARGE
SD-I	$(10, 7)$	$(10, 2)$	2	8	$(15, 10, 2)$	10×10
SD-II	$(10, 4)$	$(10, 4)$	3	8	$(15, 10, 4)$	20×20
Cri	$(10, 5)$	$(10, 2)$	3	8	$(15, 10, 5)$	20×20
Gulfport	$(10, 8)$	$(10, 3)$	3	8	$(15, 10, 8)$	10×10
Cooke	$(10, 9)$	$(10, 7)$	5	14	$(15, 10, 8)$	32×20
Synthetic scene	$(10, 2)$	$(10, 5)$	5	14	$(15, 10, 28)$	20×20

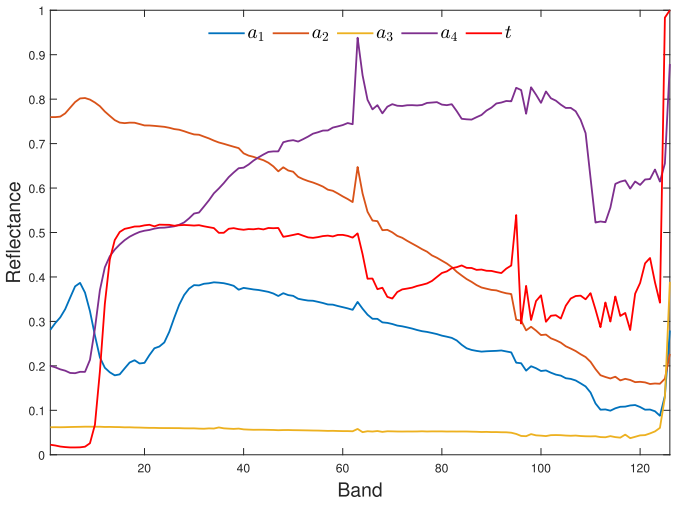

Fig. 4. Spectral signatures of four kinds of vehicles and one fabric panel.

Also, anomalies in different columns have different shapes and sizes. The sizes from the first column to fifth column are 1, 3, 4, 5, and 8, respectively. The false-color image of the synthetic scene is shown in Fig. 3(d).

B. Experimental Settings

This article concentrates on FE. To evaluate the performance of the proposed ARGE method, non-DR process and five classical DR methods, LLE, LLTSA, OTVCA, JASD, and S^3-PCA, are used for comparison. In these five DR methods, the LLE and LLTSA are graph-based DR methods.

TABLE II
PRSD OF DIFFERENT DISTANCE MEASURES

	$(\boldsymbol{a}_1, \boldsymbol{a}_2)$	$(\boldsymbol{a}_1, \boldsymbol{a}_3)$	$(\boldsymbol{a}_1, \boldsymbol{a}_4)$	$(\boldsymbol{a}_2, \boldsymbol{a}_3)$	$(\boldsymbol{a}_2, \boldsymbol{a}_4)$	$(\boldsymbol{a}_3, \boldsymbol{a}_4)$
ED	1.1909	1.1062	1.4266	1.0766	1.6990	1.5782
SAM	1.1802	1.0534	1.4256	1.1204	1.6825	1.5017
RD	1.1595	1.1403	1.7023	1.3221	1.9738	1.4929
SID	1.3520	1.3025	3.2571	1.7609	4.4036	2.5007
SAM-ED	1.3868	1.1609	1.9991	1.1946	2.7724	2.3208
SAM-SID	1.5744	1.2411	4.5641	1.9539	7.1855	3.6775
RD-ED	1.3961	1.0383	2.4789	1.4496	3.4608	2.3874
RD-SAM	1.3835	1.0904	2.4771	1.5086	3.4272	2.2717
RD-SID	**1.5849**	**1.4960**	**5.6594**	**2.3711**	**8.9698**	**3.7829**

TABLE III
COMPARISON OF DIFFERENT WINDOW SIZES

Dataset (spatial size)	Performance	Window Size			
		5×5	10×10	20×20	50×50*
SD-I	AUC	**0.9979**	0.9973	0.9972	0.9615
(100×100)	Time (s)	5.4326	0.5793	0.4010	**0.3682**
SD-II	AUC	0.9430	0.9435	**0.9525**	0.9327
(200×200)	Time (s)	91.1105	6.5613	1.5258	**1.4119**
Cri	AUC	0.9741	0.9729	**0.9781**	0.9639
(400×400)	Time (s)	1004.7049	35.7162	2.3744	**1.3379**
Gulfport	AUC	**0.9648**	0.9647	0.9631	0.9597
(100×100)	Time (s)	5.6388	0.6305	0.3942	**0.3236**
Synthetic scene	AUC	0.9840	**0.9913**	0.9875	0.9858
(280×280)	Time (s)	303.4267	15.6696	1.9905	**1.7124**

* The largest window size of synthetic scene is 56×56 instead of 50×50.

The embedding dimensions of different methods are tuned to be their best performances. The numbers of neighbors in the LLE, LLTSA, and S^3-PCA are uniformly set to be 10. As suggested in the original article [45], the spatial and range standard deviations α_s and α_r are taken as 350 and 0.4 in JASD, respectively.

As for the ARGE, an important hyperparameter is the number of subregions after partitioning. To strike a balance between performance and cost, the original HSI is segmented into several sub-HSIs via sliding window method. The impact of window size will be discussed in Section IV-D. Also, the quantity of reference vertices in the ARGE is always closely related to the amount of information contained in HSI. Taking the Cooke dataset as an example, there are two reference vertices similar to the background and another three far away from the background.

The setting of optimal parameters is presented in Table I. The two parameters in LLE and LLTSA stand for the number of neighbors and dimension. As for the OTVCA and JASD, only the dimension is required. Besides, the window size is the only parameter in the ARGE. The parameters in triple set of S^3-PCA are the number of superpixels, the number of neighbors, and dimension.

The low-dimensional feature provides the efficiency for the subsequent research. Anomaly detection is used as one of potential applications, where the state-of-the-art RX detector is adopted. This article utilizes receiver operating characteristic (ROC) curve, area under curve (AUC), and separation map to evaluate detection performance.

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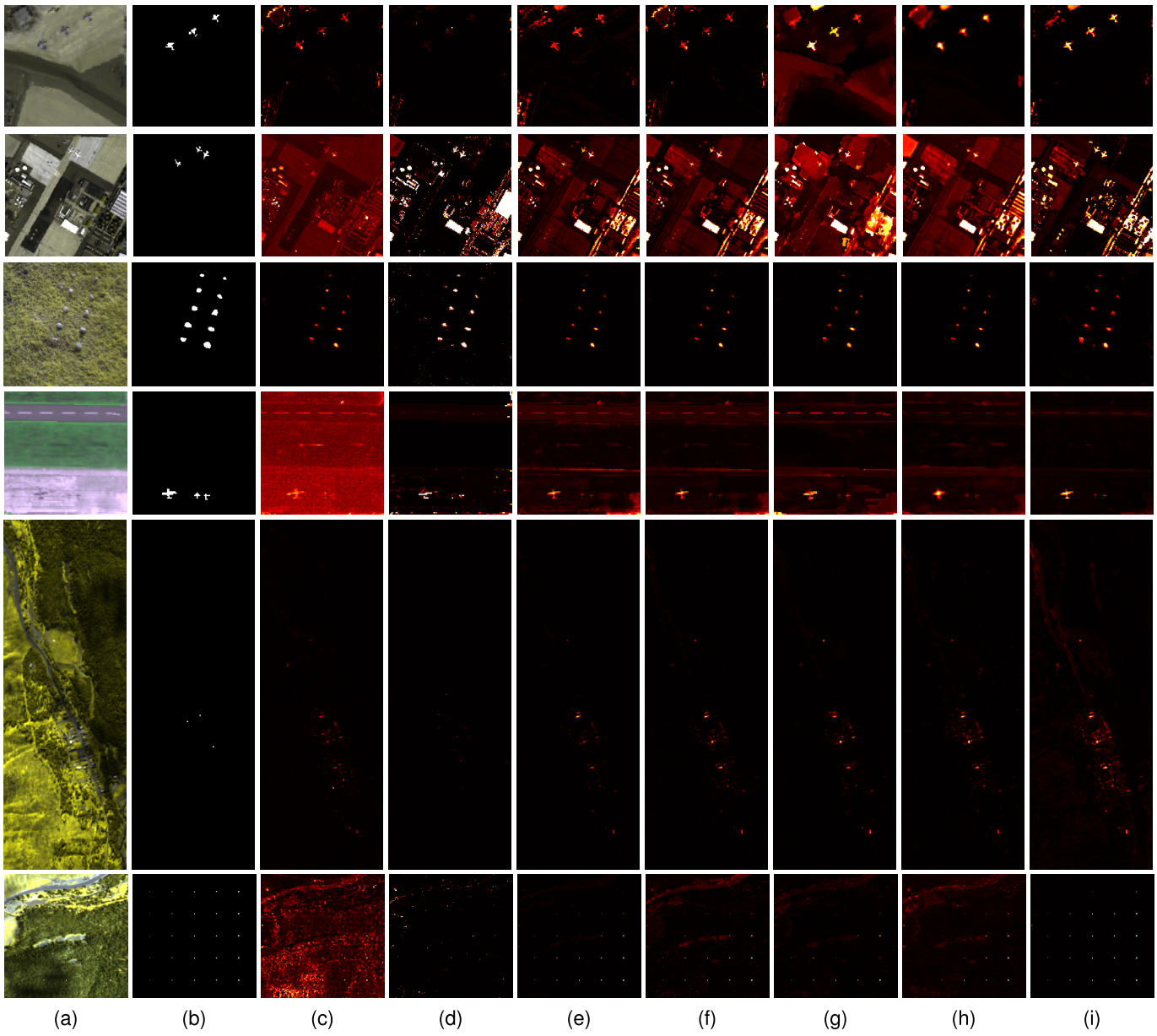

Fig. 5. Detection performances of the compared DR methods by applying RX detector. (a) False-color images of SD-I (the 1st row), SD-II (the 2nd row), Cri (the 3rd row), Gulfport (the 4th row), Cooke (the 5th row), and Synthetic scene (the 6th row). (b) Ground-truth maps. (c) Raw. (d) LLE. (e) LLTSA. (f) OTVCA. (g) JASD. (h) S^3-PCA. (i) ARGE.

C. Comparison of Distance Measures

Choosing an appropriate distance measure to reflect spectral discrepancy is an important and challenging task. Nine dissimilarity measures, including four single distances ED, SAM, RD, and SID and five hybrid ones SAM-ED, SAM-SID, RD-ED, RD-SAM, and RD-SID, are selected for comparison.

To evaluate the effectiveness of the proposed spectral measure RD-SID, a well-known criterion proposed in [72] is adopted, which is referred to as the PRSD in this article. The PRSD is given by

$$\mathrm{PRSD}(\boldsymbol{a}_i, \boldsymbol{a}_j; \boldsymbol{t}) = \max\left\{\frac{d(\boldsymbol{a}_i, \boldsymbol{t})}{d(\boldsymbol{a}_j, \boldsymbol{t})}, \frac{d(\boldsymbol{a}_j, \boldsymbol{t})}{d(\boldsymbol{a}_i, \boldsymbol{t})}\right\} \tag{20}$$

where $\boldsymbol{a}_i$ and $\boldsymbol{a}_j$ are two spectral signatures under test and $\boldsymbol{t}$ is a fixed spectral signature for referring to. The PRSD shows the efficacy of a distance measure $d(\cdot, \cdot)$ on distinguishing $\boldsymbol{a}_i$ from $\boldsymbol{a}_j$ relative to $\boldsymbol{t}$.

The experiment of PRSD selects four spectral signatures $\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$, and $\boldsymbol{a}_4$, corresponding to four kinds of vehicles "V1," "V2_back," "V2_cabin," and "V3," as basic spectra to obtain six pairwise PRSDs. The relative spectrum $\boldsymbol{t}$ corresponds to the fabric panel "F1." These five different spectral signatures are provided by the Center for Imaging Science, Rochester Institute of Technology [68] and are shown in Fig. 4.

Table II shows the pairwise PRSDs of different distance measures. It can be clearly observed that the PRSDs of hybrid-distance measures are greatly superior to that of the single-distance measures. Among all the single distances, the SID is exceedingly good and the RD ranks second. The RD-SID possesses the best spectral discrimination in all

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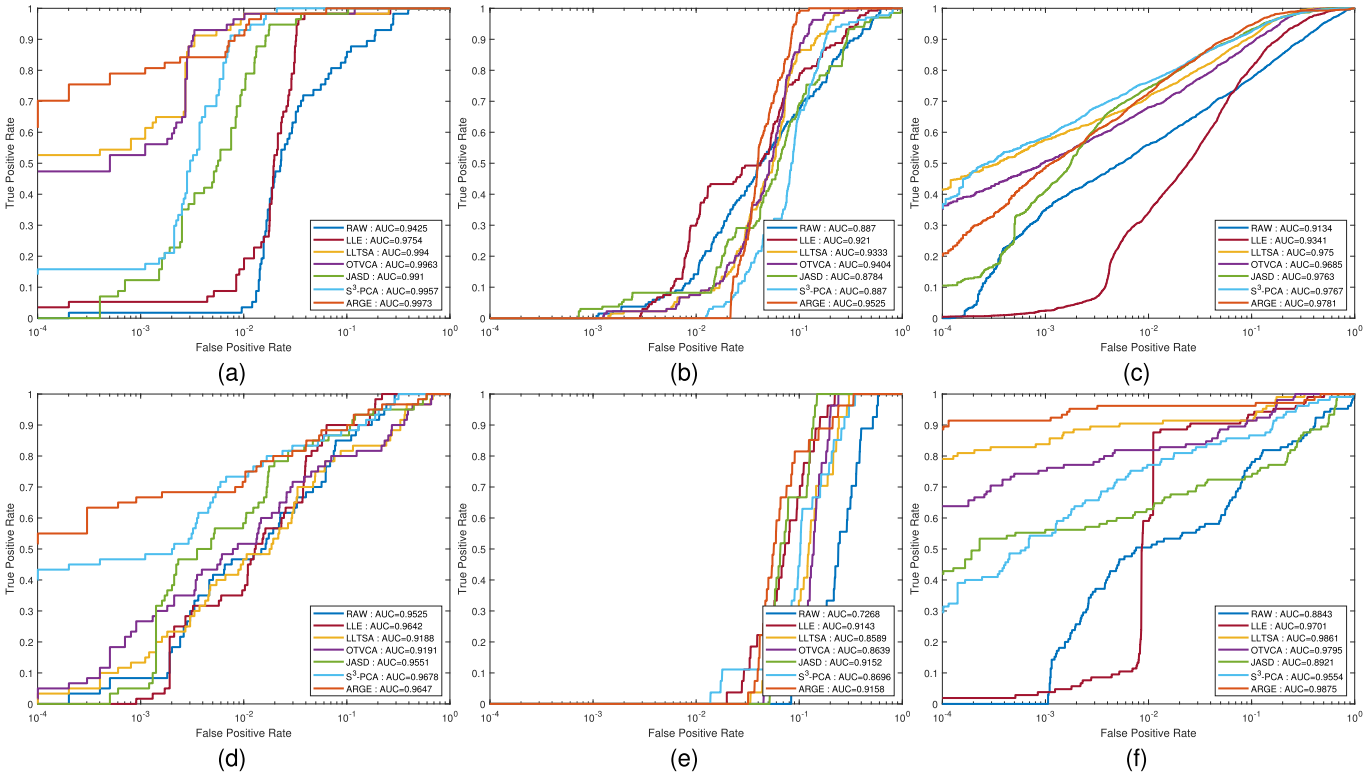

Fig. 6. ROCs and corresponding AUCs of the compared DR methods on different datasets by applying RX detector. (a) SD-I. (b) SD-II. (c) Cri. (d) Gulfport. (e) Cooke. (f) Synthetic scene.

dissimilarity measures, especially in the pairs $(\boldsymbol{a}_2, \boldsymbol{a}_4)$ and $(\boldsymbol{a}_1, \boldsymbol{a}_4)$. In other words, the RD-SID is more sensitive to distinguish $\boldsymbol{a}_4$ from $\boldsymbol{a}_1$ or $\boldsymbol{a}_2$ relative to $\boldsymbol{t}$.

D. Different Window Sizes

The basic parameter in the ARGE is the window size. To verify the influence of window size on the ARGE method, different window sizes are tested.

As shown in Table III, the variation of window size only has a slight effect on detection result. The 5×5 size brings about the best detection performance in the SD-I and Gulfport datasets. However, it is time-consuming compared with other sizes. It is sounder to choose 10×10 size in the SD-I and Gulfport. The size of 20×20 achieves the best result and saves a considerable amount of computational cost in SD-II, Cri, and synthetic scene. Therefore, the window size of 20×20 is selected in the SD-II, Cri, and synthetic scene. As for the Cooke dataset, since it is a spatially nonsquare image, several rectangular windows of different sizes are selected for comparison. Among all these sizes, 32×20 is set as a suitable window size for the Cooke dataset due to its good performance.

It is also noteworthy that there exists nonmonotonous fluctuation with respect to AUC as the window size increases. In fact, there is no theoretical evidence to guarantee the detection performance varies consistently with the window size. The size of window only has a direct impact on the number of anchor pixels and thereby influences the execution time. What really affects the detection performance is the quality of anchor pixels but not the quantity of them. In general, a small window size can preserve more local spatial information, yet it brings high parallel computing pressure. A large window size makes the embedding process faster but loses a lot of spatial information. Therefore, a moderate window width is suggested to be $n^{1/4}$, which gives a balance between performance and cost.

E. Detection Performance Analysis

In this section, the detection performance of the ARGE method is compared with typical DR methods. After obtaining the embedded feature, the RX detector is applied to appraise the effectiveness of DR methods in anomaly detection. The detection maps of five real HSIs and one synthetic scene are visually presented in Fig. 5, while the corresponding ROC curves are shown in Fig. 6. For detection maps, the brightness indicates the intensity of anomaly. The closer the detected pixel is to the anomaly, the brighter it is. Visually, the detected anomalies of the ARGE, LLTSA, OTVCA, JASD, and S^3-PCA have a strong contrast with the background in SD-I, Cri, Gulfport, and synthetic scene, while the contrast of LLE is inapparent. For the SD-II dataset, the detection visualizations of different methods are miscellaneous. The heterogeneous objects and various materials in SD-II and Cooke scenes result in the generally poor detections, including the ARGE. In addition, two superpixel-based methods JASD and S^3-PCA are relatively outstanding in detecting block-aggregated objects, such as rocks in the Cri dataset.

As for the AUCs shown in Table IV, the ARGE generates the best detection performance on different HSIs except the

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Gulfport dataset. Despite the fact that the AUC of ARGE is 0.31% lower than S^3-PCA in the Gulfport dataset, the ARGE only needs almost half execution time of S^3-PCA. Meanwhile, as shown in Fig. 6(d), the performance of ARGE is much better than S^3-PCA when the false positive rate is controlled at a low level. The LLTSA and OTVCA have high stability in the majority of HSIs but also show insufficient capacity in the Gulfport and Cooke datasets. The LLE works well in those HSIs with obvious contrast between anomaly and background, including SD-I and Gulfport. However, the LLE fails to accurately detect anomaly when assorted objects are involved in HSI. In addition, the superpixel-based methods JASD and S^3-PCA have a limitation in detecting dispersed and tiny objects since they provide relatively low AUCs in SD-II and synthetic scene. On the contrary, the proposed ARGE is more insensitive to the shape or distributed location of anomalies and is capable of capturing low-intensity abnormalities.

In terms of time consumption of DR, the ARGE and JASD show excellent computing efficiency in different image sizes compared with other DR methods. Especially for the large-spatial-size HSI like Cooke dataset, they only spend 4.3024 and 4.4701 s on DR. Quite the opposite, the traditional GE methods LLE and LLTSA are incapable of large-scaled HSI since a large amount of computational space and executing time are required. Besides, the DR processing time of OTVCA and S^3-PCA is acceptable when the HSI size is not extremely large.

In order to further analyze the separation of background and anomaly, statistical separability analysis is frequently employed [73], [74]. In this article, the separation maps of different DR methods on three real datasets are obtained by using the RX detector, which are shown in Fig. 7. The blue box stands for the normalized detection statistic of background pixels, while the red one represents the statistic of anomaly pixels. The upper and lower boundaries of the box body stand for the first and third quartiles of the normalized detection statistic, respectively. The top and bottom horizontal lines represent the maximal and minimal values, respectively. For the SD-I and Gulfport datasets, the ARGE and LLE have good suppressions of background. The rest of DR methods also have fine separations. As for the SD-II and Cooke, it is apparent that the ARGE method separates the major box of anomaly from the background. For the Cri and synthetic scene, the ARGE achieves significant separations, where there is no overlap between background and anomaly. All in all, the ARGE could enhance the power of background–anomaly separation.

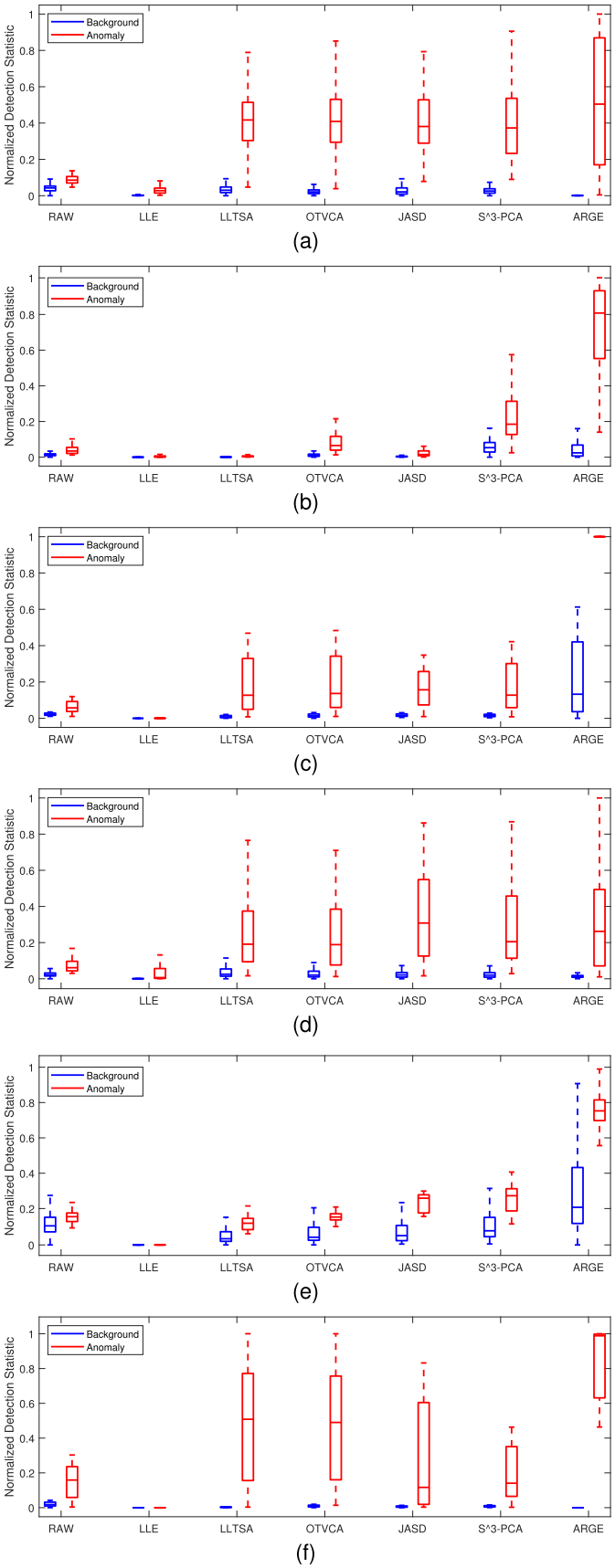

Fig. 7. Separation maps of the compared DR methods on different datasets. (a) SD-I. (b) SD-II. (c) Cri. (d) Gulfport. (e) Cooke. (f) Synthetic scene.

F. Computational Complexity

The whole process of the ARGE method can be roughly divided into four steps, which are data reduction, reference vertices selection, reference vertices supplementing, and reference-related GE. The computational complexity of $\mathcal{O}(n/m)$ is needed for downsampling, where n and m refer to the number of pixels and segmented windows, respectively. The first step selects $2m$ anchor pixels. As for the second step, constructing a similarity matrix requires $\mathcal{O}(m^2)$ operations and carrying out spectral clustering also needs $\mathcal{O}(m^2 d)$, where d is the number of clusters or the latent dimension of feature space. The supplementary step needs $\mathcal{O}(2md)$ operations.

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TABLE IV
DETECTION PERFORMANCE COMPARISON OF DIFFERENT DR METHODS BY APPLYING RX DETECTOR

Dataset (spatial size)	Performance	Method						
		RAW	LLE	LLTSA	OTVCA	JASD	S^3-PCA	ARGE
SD-I (100 × 100)	AUC	0.9425	0.9754	0.9940	0.9963	0.9910	0.9957	**0.9973**
	Time (s)	–	3.9219	10.9607	1.8138	1.5761	2.5645	**0.6514**
SD-II (200 × 200)	AUC	0.8870	0.9210	0.9333	0.9404	0.8784	0.8870	**0.9525**
	Time (s)	–	119.1958	185.0778	8.7778	2.1622	15.5811	**1.8551**
Cri (400 × 400)	AUC	0.9134	0.9341	0.9750	0.9685	0.9763	0.9767	**0.9781**
	Time (s)	–	12368.8179	7120.5498	31.9546	3.0887	155.4465	**2.6070**
Gulfport (100 × 100)	AUC	0.9525	0.9642	0.9188	0.9191	0.9551	**0.9678**	0.9647
	Time (s)	–	4.4159	11.4479	2.3383	2.2765	2.3111	**1.2597**
Cooke (800 × 280)	AUC	0.7268	0.9143	0.8589	0.8639	0.9152	0.8696	**0.9158**
	Time (s)	–	21496.8158	16659.5133	82.8516	4.4701	161.3310	**4.3024**
Synthetic scene (280 × 280)	AUC	0.8843	0.9701	0.9861	0.9795	0.8921	0.9554	**0.9875**
	Time (s)	–	2206.1568	1025.0253	24.7305	2.5546	20.3988	**1.9633**

The complexity of the final embedding step is $\mathcal{O}(nd)$. Thus, the total computational complexity of the ARGE method is $\mathcal{O}(n/m + m^2(d+1) + 2\ md + nd)$. Since $d \ll m \ll n$ in the ARGE, the complexity can be approximately scaled as $\mathcal{O}(m^2 + n)$. In particular, the ARGE method possesses linear complexity when $m^2 \approx n$ holds. For more details about the computational complexity of other algorithms, it is suggested to refer to [38].

V. CONCLUSION

In this article, an ARGE method has been proposed to extract low-dimensional features of high-dimensional data for hyperspectral anomaly detection. Constructing an HSI as a graph, some representative reference vertices are then adaptively selected by a three-step ARSS algorithm. These pivotal vertices maintain the locally sharp features of graph and eliminate redundant information. Thereafter, the ARGE embeds the high-dimensional graph into low-dimensional space through preserving the reference-related structure. Finally, the obtained features are used as the input for successive anomaly detection. The comparative experiments demonstrate that the ARGE outperforms other FE approaches.

The distinctive contribution of the ARGE method is giving an insight into a reference-related GE framework. Taking reference vertices as the core, the reference-related structure can maintain the intrinsic topology of global graph structure and remove redundant information.

A limitation of the ARGE is that the whole embedding process relies heavily on the selection of reference vertices. Also, the quality of these vertices essentially depends on the inherent topological structure of HSI. When an HSI consists of homogeneous pixels, the constructed graph is compact and the reference vertices are inseparable. Also, thereby, the poor-quality and indistinctive features are obtained. To overcome this restriction, a worthwhile extension of this work is incorporating hypergraph structure. Meanwhile, as mentioned in Section II, the embedding dimension and the number of reference vertices are considered equal in this article. It is also meaningful to extend this work to the unequal case.

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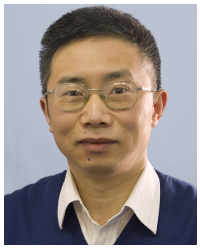

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